

CPSC 471 Project – Relational Model and Assumptions

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1 Relational Model

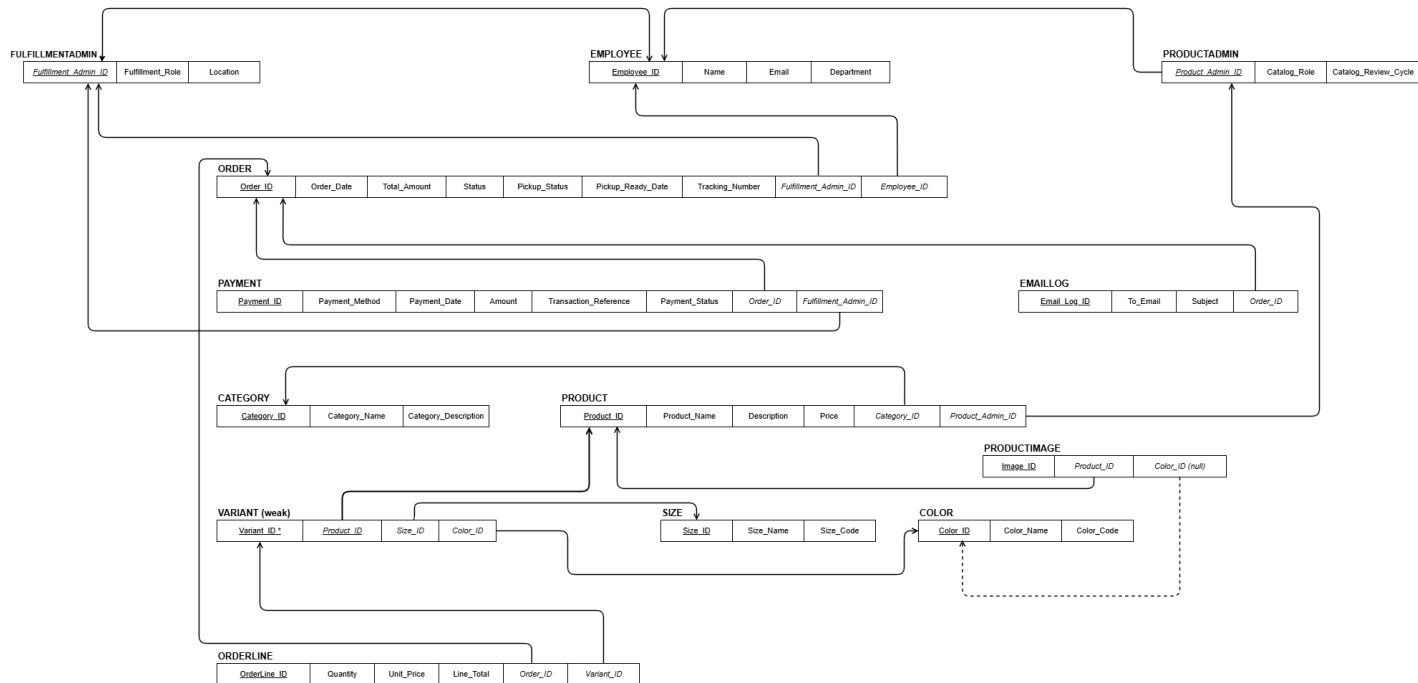


Figure 1: Relational Model Diagram (RM) for the BrightCore Merch Store system.

2 Design Updates and Key Assumptions

2.1 Design Updates Based on Feedback

- To improve role clarity, the original Admin/SuperAdmin specialization was refined into two operational subtypes of Employee: **FulfillmentAdmin** (order and payment operations) and **ProductAdmin** (catalog maintenance).
- To make responsibilities explicit in the relational design, foreign keys were added to connect role subtypes to the records they operate on:
 - `Order.Fulfillment_Admin_ID` references **FulfillmentAdmin** to indicate who fulfilled an order.
 - `Payment.Fulfillment_Admin_ID` references **FulfillmentAdmin** to indicate who recorded/handled a payment.
 - `Product.Product_Admin_ID` references **ProductAdmin** to indicate who maintains product information.
- To support clearer entity descriptions, additional descriptive attributes were included for lookup entities: `Category_Description`, `Size_Code`, and `Color_Code`.

2.2 Users and Role Specialization

- **Employee** is the supertype representing any user who can interact with the system (e.g., place orders).
- **FulfillmentAdmin** and **ProductAdmin** are subtypes of **Employee** that represent operational responsibilities.
- The specialization is **disjoint**: an **Employee** cannot be both a **FulfillmentAdmin** and a **ProductAdmin** at the same time.
- The specialization is **partial**: some **Employees** may be neither subtype (regular employees who place orders).

2.3 Catalog: Categories, Products, Images, Sizes, and Colors

- Each **Product** belongs to exactly one **Category**; a **Category** may contain zero or many **Products**.
- Each **Product** may have zero or many **ProductImages**; each **ProductImage** belongs to exactly one **Product**.
- A **ProductImage** may optionally reference a **Color** (used for color-specific images); otherwise the **Color** association is absent.
- **Size** and **Color** are modeled as lookup entities. `Size_Code` and `Color_Code` provide standardized short labels.

2.4 Variants (Weak Entity)

- **Variant** is modeled as a weak entity owned by **Product**.
- The relationship *Product identifies Variant* is an identifying relationship.
- Every **Variant** must be associated with exactly one **Product** (total participation of **Variant** in the identifying relationship).
- Each **Product** may have zero or many **Variants**.
- Each **Variant** references exactly one **Size** and exactly one **Color** in the current relational model.

2.5 Orders, Order Lines, Payments, and Email Logs

- Each Order is placed by exactly one Employee; an Employee may place zero or many Orders.
- Each Order contains one or many OrderLines; each OrderLine belongs to exactly one Order.
- Each OrderLine references exactly one Variant; a Variant may appear in zero or many OrderLines across multiple Orders.
- Each Payment belongs to exactly one Order; an Order may have one or many Payment records.
- Each EmailLog record is associated with exactly one Order; an Order may have zero or many EmailLog records.